



Week 03 - Tutorial 01



Transcript Language

English



Hello guys, today we will discuss about the subspace, what is subspace.

Suppose V is a vector space, then subspace is actually subset of vector space V with

W having some property.

So, let us go through by formal definition.

Suppose W is a subset of a vector space V over the, V is vector space over R , we call

W is a subspace of V , if W is also a vector space over R with and this vector space form



over some operation and that operation will be same whatever the operation on V is there.

Now, we know that what is subset.

So, suppose it is V is a set then some collection of element of V , if you take that then if

you form a set of those elements then we will say that set is actually a subset of V and

what is vector space?

So, we know that suppose V is a set and we define two operation addition and scalar multiplication

on V with this addition and scalar multiplication having closed properties, suppose x and y

belongs to V then if you add them, add them and if it should V belongs to V and this scalar

multiplication.

So, suppose C is a real number and x is an element of V and this x , C times V also should

belongs to V .

If suppose these properties followed in V and with as we have seen in video lecture,

there are eight properties which we use to say oximes, if those properties followed by

this operation, then we will say V is a vector space.

Now, suppose W is a subset of V and if we define the same operation addition and scalar

multiplication with the property same like this, the x belongs to W and Y belongs to

W and to making a subspace to, subspace of V it should be x plus y also belongs to W

and suppose C is a real number and x belongs to W then Cx should also belongs to W . This

is closed under addition and this is closed and this is called closed under scalar multiplication.

With same properties, this eight properties which V follows, if W will also follow then

we will say W is a subspace of V .

Now, suppose W is a given subset of V , then how we will check that W is also a subspace.

W is a subset of, it will be given then how we will check that W is subspace of V then

there is three criteria.

If W will satisfy that then we will say W is subspace of V . Those three criterias are

0 should belong to W and the second criteria is if x suppose x an element belongs to W ,

y is an element of W then addition should also so belongs to W .

This is also called closed under addition and the third criteria is, suppose c is a

real number, and x is an element of W then their multiplication which is called a scalar

multiplication and if that one, that multiplication is also element of W , which is called a closed

under scalar multiplication then these, if these criteria followed by W , then we will

say W is a subspace of V .

Now, let us do some example.

Let the given vector space is $\mathbb{R}^2$ over $\mathbb{R}$ and we have given a subset W , W is x, y coordinate

and this, if we add x plus y it will become 0 and x, y element of $\mathbb{R}$, these two conditions.

So, W is a subset given of $\mathbb{R}^2$.

We need to check W is a subspace or not.

So, let us our first criteria that 0 should belongs to W .

So, if we take x is equal to 0 and y is equal to 0, and if we add them 0 plus 0 it becomes

0, it means, 0, 0 belongs to W that means 0 element of R^2 which is also belongs to W .

Now, we need to check the second criteria, the second criteria is let W_1 is one element

is x_1, y_1 and the second W_2 second element in x_2, y_2 .

Now, if we add them w_1 plus w_2 , so it will become x_1, y_1 plus x_2, y_2 this is equal to

coordinate y addition will happen same operation, as we did in R^2 .

So, this will become the x_1 plus x_2, y_1 plus y_2 .

Now, w_1 is element of W , W_2 is element of W , it means, this two will satisfy this condition

means x_1 plus y_1 is equal to 0, and this is the for W_1 and for W_2 , X_2 plus Y_2 is equal

to 0.

So, these two conditions we have given because W_1 and W_2 two are element of W . So, now, what

we did in addition of W_1 plus W_2 we got these coordinates x_1 plus x_2, y_1 plus y_2 .

So, let us check is this addition is belongs to W , so we need to add them plus y_1 plus

y_2 we need, we added them.

And we, as we said these are the element of R^2 .

So, that swapping can we do easily so, we can write x_1 plus y_1 plus x_2 plus y_2 and this

is 0 and this is 0 as w_1 .

This is 0 as w_1 is that coordinate addition is becomes 0 as given condition and this is

0, so 0 plus 0 becomes 0.

It means, this addition belongs to W . So, second criteria is followed.

Now, we need to check third criteria.

The third criteria is a real number C and suppose W_1 is equal to x_1, y_1 an element of

W and so W_1 is element W it means x_1 plus y_1 is equal to 0.

Now, let us do multiplication C into W_1 is equal to it will become coordinate wise multiplication

will happen, same operation as in R^2 .

So, it will become Cx_1, Cy_1 .

Now we need to check this coordinates make sum 0 or not, then we will say this is scalar

multiplication is in W happened that is closed under scalar multiplication.

So, let us do the sum.

So, sum x_1 plus Cy_1 this we can write x_1 plus y_1 .

We are able to do this because as we can able to do in R^2 , so we can do this and we have

given and this is 0 it means this is 0.

It means C times W_1 is element of W . So, all three criterias done by W it means W is a

subspace of R^2 .

So, let us see it geometrically what do we mean, what we mean by a subspace.

So, this is the vector space R^2 representing this plane is actually x, y , plane R^2 means

x, y plane.

Now, we have given a subset W , which is coordinates x, y and having some condition x plus y is

equal to 0.

If you see it as in x, y plane so actually x this W represent a line, which pass through

origin with having slope minus 1.

So, actually W is this line which pass through origin and we if we see clearly the first

condition 0 it passes through origin, so 0 element is here is present in W and if we

do addition with any taking to, any two elements then it will be the same in the line and again

and the third criteria is by taking any real number.

If you multiply with this any element of W , then it will be again in the line.

So, it means W is a subspace of $\mathbb{R}^2$.

Now, suppose, another W_1 is a subspace of $\mathbb{R}^2$ and which having condition the same coordinates

with changing the slope actually.

So, subspace of $\mathbb{R}^2$ will be the passing represent the line, which passes through origin.

Why this is passed through origin because of the first criteria, 0 element would be

in W . So, any subspace of $\mathbb{R}^2$ other than trivial subspaces, which are 0 subspace and $\mathbb{R}^2$ itself,

represent a line which pass through origin.

Now, we will see a subset of $\mathbb{R}^2$, which will not form a subspace, it means, we have to

check those three criteria, which we discussed earlier and we will see that, which conditions

will not followed by this $W1$.

So, let us check this, the first criteria that 0 element in $W1$.

So, first see clearly $W1$ is actually this subset of $R2$ with the same condition, as we

discussed earlier x plus y is equal to 0, but that x is that x and y belong, taken from

this integer that means x, y are integer.

So, we know that 0 belongs to z , so we can take the coordinate 0, 0 and this follow the

condition 0 plus 0 which is 0.

So, it means 0 element is in W . So, first criteria satisfied $W1$, now we need to check

second criteria.

The second criteria is, you will take two element from $W1$.

Let us take $x1, y1$ and second element is the $x2, y2$.

So, if we from $W1$ if we add them this will become coordinate wide addition and addition

of two integer is integer.

Also previously we have seen that addition of coordinates that is the $x1$ plus $x2$ plus

$y1$ plus $y2$ is equal to 0, here, same thing will also happen.

So, conditions of W_1 are satisfied, it means it follows the addition, this W_1 is closed

under addition.

So, it means this belongs to W_1 .

So second criteria followed by W_1 .

Now, the third criteria is, any real number multiplied with an element, which is at a

scalar multiplication and this is lacking W_1 .

So, let us take any real number we can take.

So, let us take any $1/2$, this is a rational number and an element of W_1 which is we can

take $1, n$ minus 1 .

So, this is from W_1 satisfying this condition and also these are the integers.

So, this is the element of W_1 .

Now, if we multiply them, so this is, this will be a coordinate wise multiplication,

so this will become $1/2$ minus $1/2$.

Now, here is the missing part that this is not an integer, this is not an integer, it

means W_1 and is not closed under scalar multiplication.

So, here the third condition is lacking for W_1 .

So, the W_1 is a subset, but not a subspace.